

Time- and Frequency-Domain Gain Selection for PD and PID Control of a DC Motor-Angle Loop

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Abstract

Proportional–derivative (PD) and proportional–integral–derivative (PID) control remain the default choice for positioning loops, yet gains are often chosen by trial and error without a clear account of what each term does to the closed loop. This paper analyses four controllers designed for a DC motor-angle loop modelled as $G(s) = 3/(0.1s^2 + s)$: an initial PD design, two retuned PD designs and a PID design. Step-response metrics, closed-loop pole locations, loop-gain frequency response and input-disturbance rejection are computed for each. Retuning the PD gains from $K_p = 7$, $K_d = 0.063$ to $K_p = 38$, $K_d = 2.28$ raises the damping ratio of the closed-loop poles from 0.41 to 1.16, cuts the overshoot from 24.6% to 4.1%, shortens the 2% settling time from 571 to 126 ms and raises the phase margin from 44.4° to 84.7° . The residual overshoot of the overdamped design is traced to the controller zero at $-1/T_d$. Adding $K_i = 4$ leaves the reference response almost unchanged because it creates a slow pole–zero dipole near $s = -0.135$; it does remove the steady-state error to a constant load, but only on a time scale of about 29 s. A sweep of K_i quantifies this trade-off, and a single-op-amp realization reproduces the PID gains within 1.3%. **Keywords:** PID control, PD control, DC motor, pole placement, frequency response, disturbance rejection, op-amp controller

1. Introduction

The large majority of industrial control loops use some form of PID control [1]. Its three terms have simple interpretations: the proportional term sets how strongly the loop reacts to the present error, the derivative term reacts to the rate of change of the error and adds damping, and the integral term accumulates past error so that constant offsets are removed [2, 3]. These descriptions are qualitative, however, and in practice gains are often adjusted by trial and error until a response “looks good”. This paper makes the effect of each gain quantitative for a specific and common case: holding a DC motor at a commanded angle.

Classic tuning rules such as the Ziegler–Nichols ultimate-gain method [4] need a loop that reaches sustained oscillation under proportional control. The second-order, type-1 plant studied here never does: its phase never reaches -180° , so its gain margin is infinite and the ultimate gain does not exist. Gains must therefore be chosen from an explicit model. The contributions of this paper are:

- a closed-form account of how K_p and K_d move the closed-loop poles and the controller zero, and why the retuned PD design is both faster and better damped (Section 4.2);
- a time- and frequency-domain comparison of four designs, including phase margin, bandwidth and peak sensitivity (Sections 4.1 and 4.3);
- an explanation, using the pole–zero dipole created by a small K_i , of why integral action barely changes the reference response but is needed for load rejection, together with a K_i trade-off study (Section 4.4);

Technical note, October 2026. It extends the analysis I carried out for a four-person team project in the Princeton University online research program supervised by Prof. Michael Littman (February–March 2025). The extended analysis and text were prepared with AI assistance; see the Acknowledgment.

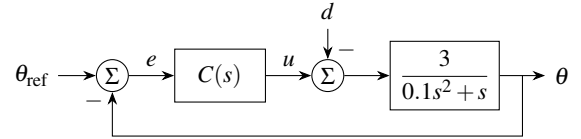


Figure 1. Motor-angle loop with controller $C(s)$, plant $G(s)$ and input (load) disturbance d .

- a check that a single-op-amp circuit realizes the PID gains (Section 4.5).

2. System Model

The motor angle θ responds to the control voltage u through

$$G(s) = \frac{\Theta(s)}{U(s)} = \frac{3}{0.1s^2 + s} = \frac{30}{s(s + 10)}, \quad (1)$$

a first-order speed response with a 0.1 s mechanical time constant followed by an integrator from speed to angle. The controller acts on the error $e = \theta_{\text{ref}} - \theta$,

$$C(s) = K_p + \frac{K_i}{s} + K_d s = K_p \left(1 + \frac{1}{T_i s} + T_d s \right), \quad (2)$$

with $T_i = K_p/K_i$ and $T_d = K_d/K_p$. A constant load is modelled as an input disturbance d that subtracts from u (Fig. 1).

With $K_i = 0$ the closed-loop transfer function is second order,

$$\frac{\Theta}{\Theta_{\text{ref}}} = \frac{30(K_d s + K_p)}{s^2 + 10(1 + 3K_d)s + 30K_p}, \quad (3)$$

so the natural frequency and damping ratio are

$$\omega_n = \sqrt{30K_p}, \quad \zeta = \frac{10(1 + 3K_d)}{2\sqrt{30K_p}}, \quad (4)$$

Table I. Controller gains studied

Design	K_p	K_d (s)	K_i (s^{-1})	T_d (s)
Initial PD	7	0.063	0	0.009
Intermediate PD	21	1.26	0	0.060
Tuned PD	38	2.28	0	0.060
PID	30	2.2	4	0.073

Table II. Closed-loop poles and step-response metrics

Design	Poles (s^{-1})	t_r (ms)	t_s (ms)	M_p (%)
Initial PD	$-5.9 \pm 13.2j$	101	571	24.6
Intermediate PD	$-23.9 \pm 7.7j$	44	184	5.0
Tuned PD	$-59.1, -19.3$	26	126	4.1
PID	$-61.4, -14.5, -0.13$	30	109	2.4

and the controller contributes a zero at $s = -1/T_d$. Equation (4) already shows the roles of the two gains: K_p alone raises ω_n and lowers ζ , while K_d raises ζ without changing ω_n . With $K_i > 0$ the loop becomes third order with characteristic polynomial $0.1s^3 + (1 + 3K_d)s^2 + 3K_p s + 3K_i$.

3. Methods

Table I lists the four designs. The initial PD gains were the team's starting point; the intermediate and tuned PD gains came from iterative tuning in circuit simulation; the PID gains were chosen to keep the speed of the tuned PD design while adding integral action. Both retuned PD designs keep the same derivative time, $T_d = 0.06$ s.

For each design I computed the unit-step response and its 10–90% rise time t_r , 2% settling time t_s and percent overshoot M_p ; the closed-loop poles and zeros; and, from the loop gain $L(s) = C(s)G(s)$, the gain-crossover frequency ω_c , phase margin (PM), closed-loop -3 dB bandwidth ω_{BW} and peak sensitivity $M_s = \max_{\omega} |1/(1 + L(j\omega))|$. Load rejection was assessed from the response of θ to a unit step in d ,

$$\frac{\Theta}{D} = \frac{-3s}{0.1s^3 + (1 + 3K_d)s^2 + 3K_p s + 3K_i}. \quad (5)$$

The original analysis was done in GNU Octave with the control package [5]; all numbers and figures in this paper were regenerated from the same model in Python with SciPy [6]. The controllers are ideal (no derivative filter or output saturation) unless stated otherwise.

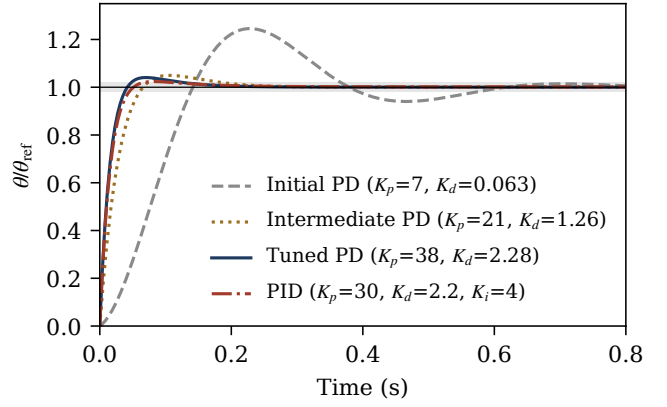
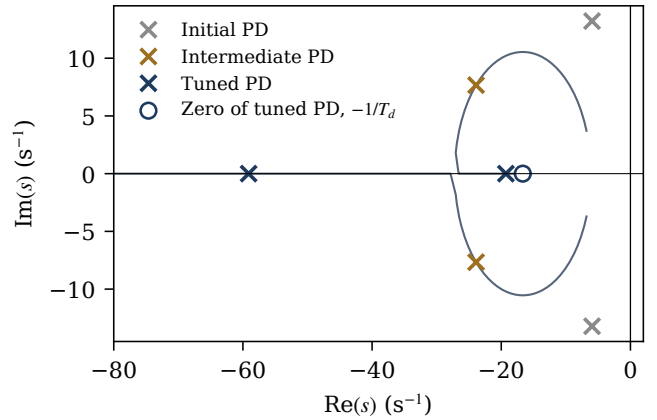
4. Results

4.1 Step response

Figure 2 and Table II compare the four designs. The initial PD loop is underdamped ($\zeta = 0.41$) and overshoots by 24.6%. Raising K_d by a factor of 20 while tripling K_p gives the intermediate design, which settles 3.1 times faster with 5.0% overshoot. The tuned PD design is faster again, with a 26 ms rise time and 126 ms settling time. The PID design has the lowest overshoot, 2.4%, and the shortest settling time, 109 ms.

4.2 Pole locations and the controller zero

Figure 3 shows how the closed-loop poles move as K_p increases with T_d held at 0.06 s. For small K_p the poles form

**Figure 2.** Unit-step responses of the four designs. The shaded band marks $\pm 2\%$ of the final value.**Figure 3.** Closed-loop pole paths for K_p from 2 to 60 with $T_d = 0.06$ s (gray), the three PD designs ($\times$) and the zero of the tuned design ($\circ$).

a lightly damped complex pair; at $K_p \approx 25$ they meet on the real axis and then split, one moving left without limit and one approaching the zero at $-1/T_d = -16.7$. The tuned design has real poles at $-59.1, -19.3$, so by (4) it is overdamped ($\zeta = 1.16$). It still overshoots by 4.1% because the controller zero at -16.7 lies to the right of the slow pole at -19.3 : a zero closer to the origin than the dominant pole adds a derivative-like component to the response that a pole-only model would miss [2]. Table III quantifies the sweep: above $K_p \approx 30$ further gain mainly shortens the rise time, while the overshoot stays near 3–5% and the slow pole stays near -18 .

4.3 Frequency response

Table IV lists the loop-gain margins. Because $\angle G(j\omega)$ only approaches -180° and the PD zero adds phase lead, every design has an infinite gain margin; robustness is therefore judged by phase margin and peak sensitivity. The initial design has a 44.4° phase margin and $M_s = 1.55$, which means that disturbances near crossover are amplified by about 55%. The retuned designs have phase margins above 80° and $M_s = 1.00$, and their crossover frequency and bandwidth are about three to five times higher. Figure 4 compares the tuned PD and PID loop gains. They are nearly identical above 1 rad/s; below about 0.5 rad/s the integrator in the PID loop adds

Table III. Effect of K_p with $T_d = 0.06$ s ($K_i = 0$)

K_p	K_d	Poles (s^{-1})	t_r (ms)	t_s (ms)	M_p (%)
5	0.30	$-9.5 \pm 7.7j$	143	412	3.7
7	0.42	$-11.3 \pm 9.1j$	108	346	4.6
10	0.60	$-14.0 \pm 10.2j$	81	284	5.2
15	0.90	$-18.5 \pm 10.4j$	58	225	5.3
21	1.26	$-23.9 \pm 7.7j$	44	184	5.0
30	1.80	$-43.1, -20.9$	32	147	4.5
38	2.28	$-59.1, -19.3$	26	126	4.1
50	3.00	$-81.6, -18.4$	21	102	3.6
60	3.60	$-100.0, -18.0$	18	88	3.2

Table IV. Loop-gain and closed-loop frequency metrics

Design	ω_c (rad/s)	PM ($^\circ$)	ω_{BW} (rad/s)	M_s
Initial PD	12.9	44.4	19.9	1.55
Intermediate PD	39.8	81.4	45.0	1.00
Tuned PD	69.6	84.7	75.6	1.00
PID	66.6	87.0	70.0	1.00

20 dB/decade of gain and pushes the phase toward -180° . Integral action is therefore a low-frequency correction that leaves the crossover region, and hence the transient response, almost untouched.

4.4 Role of integral action

For reference tracking the plant already contains an integrator, so every design in Table I reaches the commanded angle with zero steady-state error. The benefit of K_i appears only under load. For a constant input disturbance d , (5) gives a steady-state angle error of d/K_p for PD control and zero for PID control. With $K_i = 4$, however, the PID loop has a pole at $s = -0.135$ almost on top of a zero at $s = -0.135$. This dipole hardly affects the reference response, which explains why $K_i = 4$ made little visible difference in simulation, but it sets the speed at which load error is removed: the time constant is about $1/0.135 \approx 7.4$ s, and the error takes 29.4 s to fall within 2% of its peak (Fig. 5).

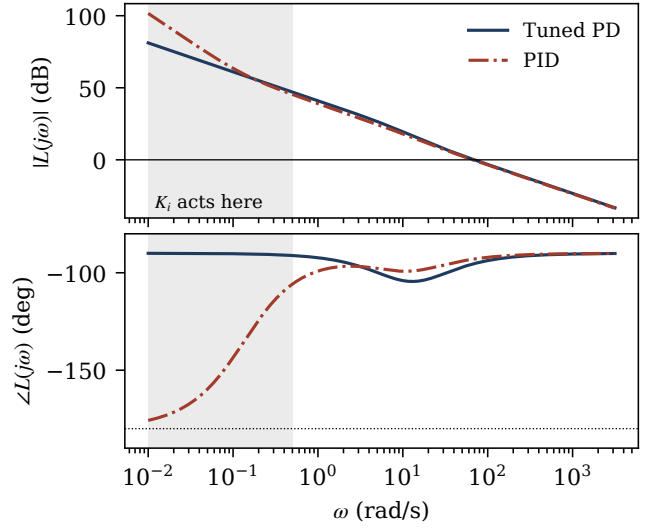
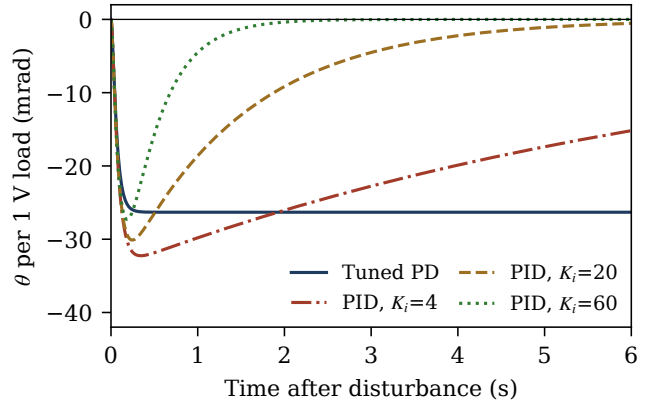
Table V shows the trade-off as K_i increases with $K_p = 30$ and $K_d = 2.2$. Raising K_i to 20 removes load error within 5.8 s while keeping the reference overshoot below 3%; $K_i = 60$ removes it within 1.8 s but more than doubles the reference settling time because the dipole moves toward the dominant poles. For a positioning loop that must hold its angle against a load, a value near $K_i = 20$ is a better compromise than the original $K_i = 4$.

4.5 Op-amp realization

The team implemented the PID controller with a single inverting op-amp stage whose input branch is R_1 in parallel with C_1 and whose feedback branch is R_2 in series with C_2 [7]. Its gains are

$$K_p = \frac{R_2}{R_1} + \frac{C_1}{C_2}, \quad K_i = \frac{1}{R_1 C_2}, \quad K_d = R_2 C_1. \quad (6)$$

Table VI shows that the component values used in the circuit reproduce the target gains within 1.3%. The cross term C_1/C_2

**Figure 4.** Bode plot of the loop gain $L(j\omega)$ for the tuned PD and PID designs. The shaded region is where K_i changes the loop.**Figure 5.** Angle deviation after a 1 V step load disturbance. PD control leaves a permanent offset; PID control removes it at a rate set by K_i .

adds 0.3 to K_p , a reminder that the three gains of a single-stage circuit cannot be set independently.

5. Discussion

Three limitations bound these conclusions. First, the derivative terms are ideal. A real differentiator needs a low-pass filter, which adds phase lag near $\omega = N/T_d$ for filter ratio N and will lower the phase margins of Table IV; a filter with $N \geq 10$ keeps that lag above the tuned crossover of about 70 rad/s. Second, the ideal derivative produces an impulse in u at a reference step. In the circuit simulations this appeared as op-amp saturation, which the team removed by lowering the supply voltage; applying the derivative to the measured angle instead of the error would avoid the kick altogether. Third, the plant model in (1) has no friction, backlash or amplifier dynamics. Because Ziegler–Nichols tuning was not applicable to this model, the gains should be confirmed on hardware, where those effects could produce the sustained oscillation that the model rules out.

Table V. Trade-off in K_i ($K_p = 30$, $K_d = 2.2$). $|p_{\text{slow}}|$: slowest closed-loop pole; e_{pk} , e_{ss} : peak and steady error per 1 V load; t_d : 2% recovery time.

K_i	$ p_{\text{slow}} $	t_s (ms)	M_p (%)	e_{pk} (mrad)	e_{ss} (mrad)	t_d (s)
0	14.676	104	2.3	33.3	33.3	—
4	0.135	109	2.4	32.3	0	29.4
20	0.709	136	2.8	30.1	0	5.8
60	2.517	263	3.8	27.3	0	1.8

Table VI. Single-op-amp PID realization

Quantity	Circuit value	Target
R_1, R_2	1.1 k Ω , 33.1 k Ω	—
C_1, C_2	67 μF , 225 μF	—
K_p from (6)	30.39	30
K_i (s^{-1})	4.04	4
K_d (s)	2.218	2.2

6. Conclusion

For the motor-angle loop $G(s) = 3/(0.1s^2 + s)$, the main effect of retuning the PD controller was to raise the damping ratio from 0.41 to 1.16 while more than doubling the natural frequency, which cut the overshoot from 24.6% to 4.1% and the settling time by a factor of 4.5. The remaining overshoot comes from the controller zero, not from the poles. Integral action with $K_i = 4$ adds a slow pole-zero dipole that leaves the step response almost unchanged; it removes the steady-state error to a constant load, but only over tens of seconds, and $K_i \approx 20$ would give a better balance between load rejection and transient speed. Future work will add a derivative filter and actuator limits to the model and validate the gains on a motor test bench.

Acknowledgment

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